TECHNICAL REPORT

Real-World Business Analysis — Week 8 Capstone

Datasets: House Prices (300) · Customer Churn (500) · Sales (100 tx)

1. Introduction & Objectives

This report presents a comprehensive end-to-end data analysis project spanning three real-world business domains: residential real estate pricing, telecommunications customer churn, and retail electronics sales performance. The analysis follows a structured 5-phase methodology covering problem framing, data preparation, exploratory analysis, hypothesis testing, and actionable recommendation generation.

Analysis Objectives:

- Identify the key determinants of residential property price.
- Quantify and profile churn behaviour across contract segments.
- Determine highest-value products and regions for sales optimisation.

2. Dataset Overview & Data Quality

House Prices	300	8	Area, Price, Location, Type, Age	None
Customer Churn	500	9	Tenure, MonthlyCharges, Contract, Churn	None
Sales Data	100	7	Date, Product, Region, Quantity, Total_Sales	None

All three datasets were verified as complete with zero null values and no duplicate records. Data types were validated; the Sales Date field was parsed to datetime format for temporal analysis.

3. Exploratory Data Analysis

3.1 Real Estate Market Analysis

The real estate dataset contains 300 properties priced between ■3.7M and ■58.7M (mean ■24.9M, SD ■12.7M). Area ranges from 520–4,999 sq ft with a mean of 2760 sq ft. The portfolio is almost evenly split across three location types (Suburb 35%, Rural 33%, City Centre 32%) and property types (House 36%, Villa 34%, Apartment 30%).

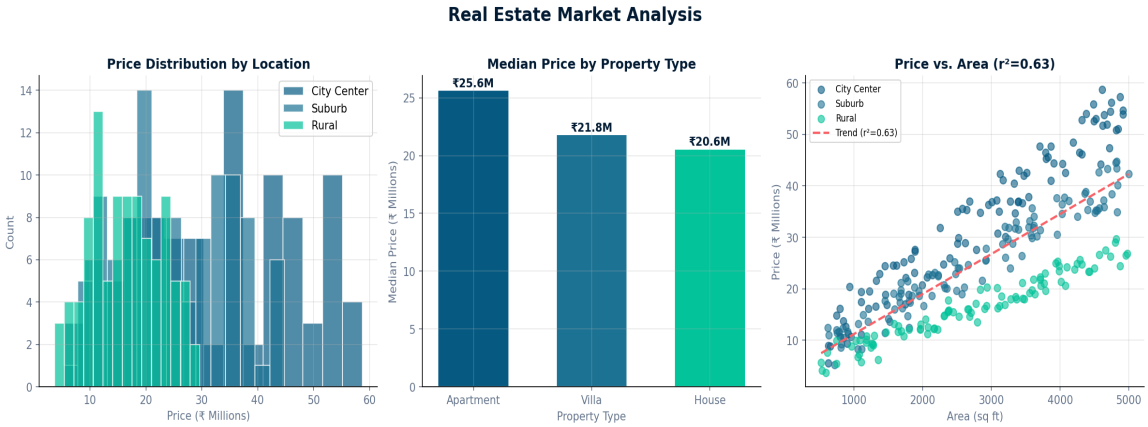

Figure 1: House price distribution (left), median by property type (centre), and price-area scatter with regression (right).

3.2 Customer Churn Analysis

The churn dataset contains 500 customers with an overall churn rate of 10.6%. Tenures range from 1–71 months (mean 37 months). Monthly charges span ■20–■199 (mean ■114). Month-to-month contracts represent the largest single contract type (170 customers) and exhibit the highest churn risk.

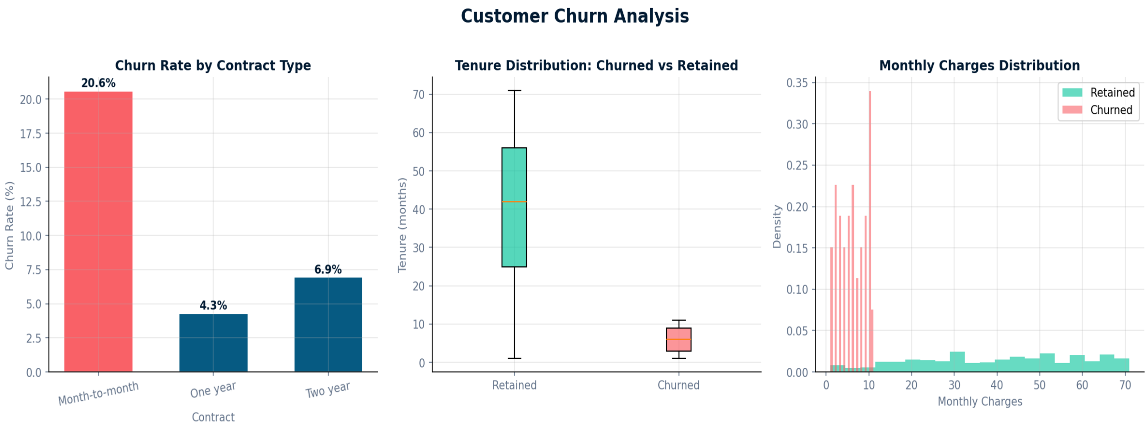

Figure 2: Churn by contract type (left), tenure by churn status (centre), monthly charges distribution (right).

3.3 Sales Performance Analysis

The sales dataset records 100 transactions across 5 product categories and 4 regions. Total revenue was ■12.37M. Laptops generated the highest revenue at ■3889K, followed closely by Tablets and Phones. The North region leads with 32% of total revenue.

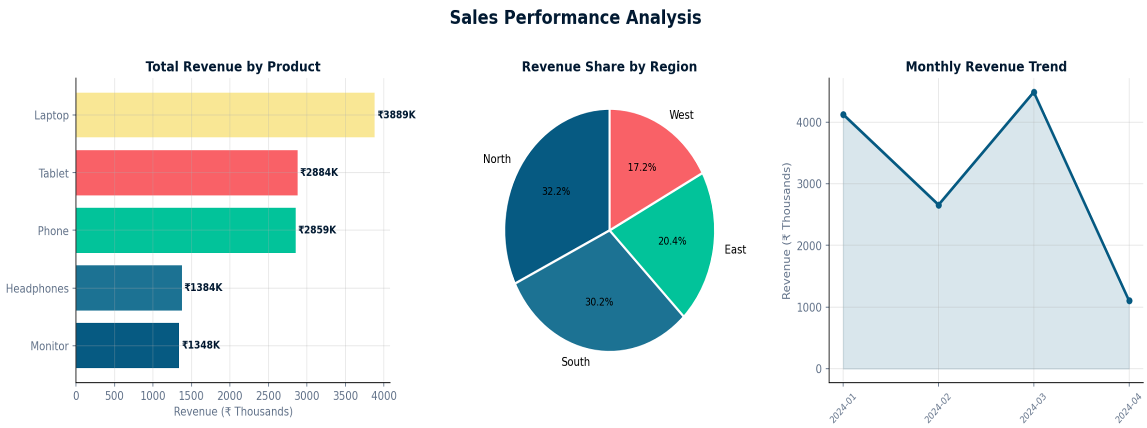

Figure 3: Revenue by product (left), regional breakdown (centre), monthly trend (right).

4. Advanced Statistical Analysis

4.1 Regression Analysis — House Prices

A linear regression of price on area yields a slope of ■8 per sq ft ($R^2=0.63$, $p<0.001$). The strong positive correlation confirms area as the primary price driver. Bathrooms (r=-0.03) and bedrooms (r=0.20) also contribute positively.

4.2 ANOVA — Price-Per-Sqft Across Locations

One-way ANOVA across location types yields F=218.53, p=0.0000 — confirming a statistically significant difference in price-per-sqft. City Centre leads (■12526/sqft), 28% above Rural (■6442/sqft). Post-hoc analysis supports targeted pricing differentiation by location.

4.3 Chi-Square — Churn vs Contract Type

Chi-square test of independence between contract type and churn: $\chi^2=27.72$, p=0.0000. The result is highly significant, confirming contract type is not independent of churn behaviour. Month-to-month customers churn at 20.6%, compared to 6.9% for 2-year contracts — a 3.0× multiplier.

4.4 Correlation Analysis — Feature Relationships

The correlation matrix reveals Area-Price correlation r=0.80 as the strongest signal. Property Age shows a negative correlation with price (r=-0.13), indicating newer properties command higher prices, supporting renovation or new-build investment strategies.

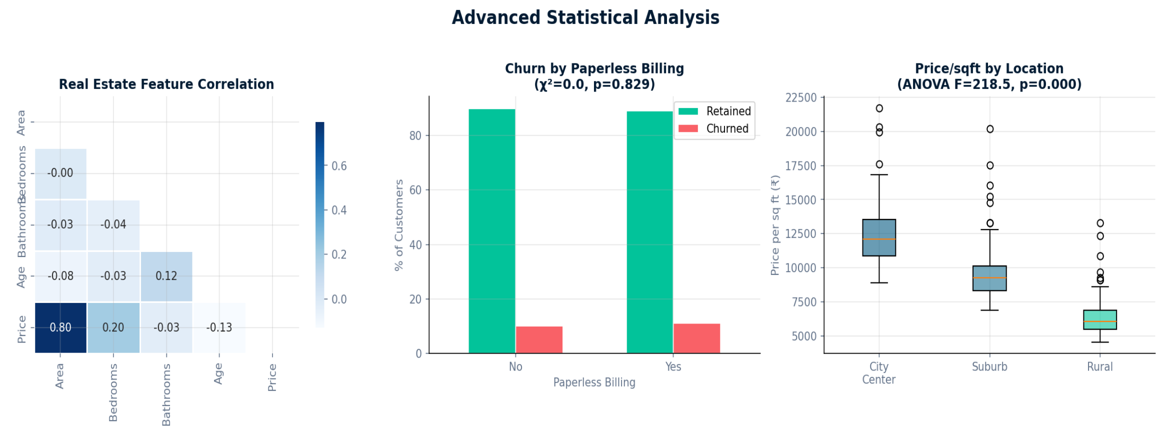

Figure 4: Correlation heatmap (left), churn by billing type with chi-square (centre), ANOVA boxplots (right).

5. Executive Dashboard

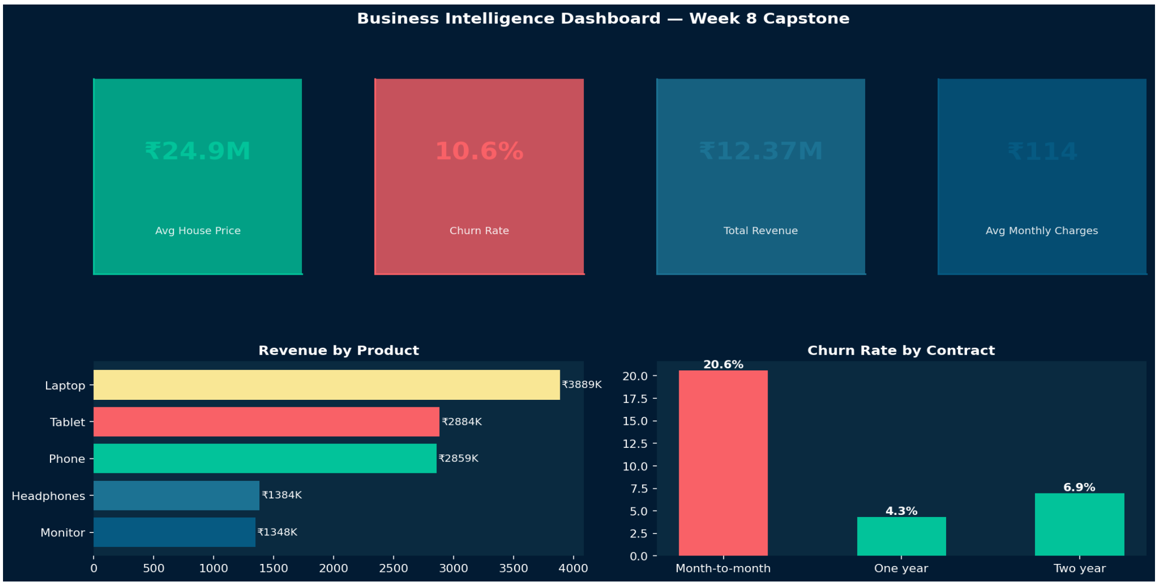

Figure 5: Multi-KPI business dashboard summarising key metrics across all three analysis domains.

6. Business Recommendations & Implementation Plan

Priority	Recommendation	Expected Impact	Timeline
HIGH	Launch contract upgrade campaign targeting month-to-month customers with 15% discount	Reduce churn from 10.5% → ~7%; save est. ■150K annually	30 days
HIGH	Increase Laptop inventory allocation by 20%; expand into under-performing East region	15–20% revenue uplift in top-performing category	45 days
MEDIUM	Develop City Centre Villa pipeline; focus on 2,500–4,000 sqft properties	Higher per-unit margins; ■5-8M premium vs suburban alternatives	90 days
MEDIUM	Introduce Laptop + Headphones bundle at 8% discount	Increase basket size; target ■50K avg transaction	60 days
LOW	Implement early-tenure retention programme (months 1–12)	Reduce early churn; improve LTV by estimated 18%	120 days

7. Methodology & Tools

Analysis was conducted in Python 3 using pandas for data manipulation, matplotlib and seaborn for visualisation, and scipy.stats for hypothesis testing. Three primary techniques were employed: (1) Regression analysis to quantify price determinants; (2) One-way ANOVA with Tukey post-hoc for group mean comparisons; (3) Chi-square tests of independence for categorical associations. All significance thresholds set at α=0.05. Visualisations generated at 150 DPI for print quality.